

DURGA PAVAN KUMAR PAILLA

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Summary

Software Engineer with 3+ years of professional experience building and supporting backend and full-stack applications in enterprise environments. Strong background in Java and Spring Boot for designing secure, scalable REST APIs, with hands-on experience deploying and operating services on AWS. Comfortable working across the stack with React and modern frontend tools, and experienced in integrating databases, caching, messaging systems, and automated testing into production-ready systems. Proven ability to collaborate in Agile teams and deliver reliable features end-to-end.

Experience

Travelers

May 2024 – Present

Software Engineer

United States

- Designed and developed a scalable **Spring Boot** Spring Boot microservices handling 500+ TPS for a personal insurance platform serving 2M+ policyholders, ensuring 99% availability during peak enrollment windows.
- Developed event-driven serverless functions using **Node.js** and **AWS Lambda** for asynchronous policy update processing, reducing manual backend reconciliation effort by **35%**.
- Optimized high-volume REST APIs by implementing **Redis** caching and database indexing, reducing P99 latency by **40%** and improving adjuster workflow efficiency.
- Implemented **OAuth2** and **JWT**-based authentication using Spring Security to enable secure, role-based access across backend services.
- Developed a scalable audit logging service using Amazon **DynamoDB** to track 1M+ daily sensitive user events with 10ms retrieval latency, ensuring strict adherence to SOC2 and regulatory compliance standards.
- Integrated **AWS KMS** encryption for sensitive policyholder data to improve data security and meet enterprise compliance standards.
- Contributed to frontend enhancements using **React**, Redux Toolkit, and Material UI, improving usability and accessibility of insurance applications.
- Wrote and maintained unit and integration tests using **JUnit** and **Mockito**, improving service reliability and reducing regression issues during releases.

Sogeti (Capgemini)

Jan 2022 – Jun 2023

Software Developer

Hyderabad, India

- Developed Java Spring Boot **REST APIs** for booking and reporting modules within a microservices-based application.
- Integrated backend services with **React** and **TypeScript** frontends to enable seamless data flow between UI and APIs.
- Designed and implemented **Redis**-based caching strategies for read-heavy APIs, reducing database load and improving response times under peak traffic.
- Worked with **PostgreSQL** to optimize queries and improve response times for core application workflows.
- Supported **AWS** infrastructure tasks including EC2, S3, and IAM configuration, and monitored applications using CloudWatch.
- Automated **CI/CD** pipelines using GitHub Actions and Docker, enabling consistent and reliable deployments.
- Added automated tests and participated in code reviews to maintain code quality across backend and frontend components.

Education

Master of Science in Computer Science

Dec. 2024

University of Central Missouri

Warrensburg, MO

Bachelor of Technology in Electronics and Communication Engineering

May 2023

National Institute of Technology Andhra Pradesh

India

Technical Skills

Languages: Java 17/21, JavaScript, TypeScript, Python, SQL

Backend & Frameworks: Spring Boot, Spring Security, Hibernate/JPA, Node.js, REST APIs, OAuth2, JWT

Frontend: React, Redux Toolkit, Next.js, Material UI, Angular

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, KMS, SQS), Docker, Kubernetes, Terraform, CI/CD

Messaging & Observability: Kafka, RabbitMQ, Prometheus, CloudWatch

Testing & Tooling: JUnit, Mockito, Git, GitHub Actions, Postman, Swagger

Projects

Sentinel – Distributed Event Monitoring System | *Java, Spring Boot, Kafka, Redis, PostgreSQL, Docker*

- Built a distributed backend system using Spring Boot microservices and Kafka to process high-volume event streams with low-latency ingestion.
- Implemented Redis-based sliding window logic and asynchronous persistence to PostgreSQL for real-time anomaly detection and auditing.

Resilient API Gateway and Rate Limiter | *Spring Cloud Gateway, Redis, Resilience4j, Prometheus*

- Developed a custom API gateway with distributed rate limiting using Redis to protect backend services from traffic spikes.
- Integrated circuit breaker patterns with Resilience4j and exposed metrics via Prometheus to prevent cascading failures.

Extracurricular

Publications: Enhanced Physical Layer Security for RIS-aided Wireless Networks (IEEE)

Certifications: AWS Certified Solutions Architect – Associate; AWS Certified AI Practitioner